3 Evolution of Mind

3.1 Intro.

The Piths had been preparing for this day for nine months, even longer considering the planning they had done to finally get Mrs. Pith pregnant. It was their first, with hopefully many to come. Nothing could spoil their excitement for those first few moments holding their baby.

But they would have to wait a bit longer. After a relatively quick labor, the nurse took the infant out of the room and returned with a lawyer, and a different child. The new child had been born at the same time, and with a little tweaking of a few genes would even grow up to look like his soon-to-be parents, but was made of the egg and sperm of separate people. The baby boy, that the Pith's had never seen, was given to the new mom, and the nurse introduced the lawyer as he opened his briefcase. "Mr. Solon has a few things for you to sign, then you can head home with your child." She lightly caressed the baby's head. There were just a few hairs on his head, but they were already black just like Mr. Pith's.

The lawyer handed Mr. Pith some documents. "As you both know, you have elected to join CAS-9's new foster program. I know you know all of this already from the pre-natal meetings, but I must go over this again, company policy." The two parents nodded but were looking at their son. The lawyer continued,

"You have agreed to relinquish any and all legal rights to the infant you have just given birth to, in exchange for your son you are currently holding. Additionally, CAS-9 will pay for all hospital fees for up to five days and the agreed upon projected cost of child rearing until they come of age. This contract is non-negotiable. Once you accept here today, you will not have any communication, written or spoken, with the infant in the other room. The infant will be placed with another family. You will not have any visitation rights, nor will CAS-9 provide you with any information regarding the whereabouts of said infant. Genetic testing via third-party providers will not provide any information. The infant's DNA will be altered to test negatively against your own, and your son you are currently holding has already been altered

Anthology of Mind

Eoin O'Hearn

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edge of chaos. So, while I imagine the trajectory of my life after graduation, I find solace in the words of Thoreau and Crighton.

> "'But,' says one, 'you do not mean that the students should go to work with their hands instead of their heads?' I do not mean that exactly, but I mean something which he might think a good deal like that; I mean that they should not play life, or study it merely, while the community supports them at this expensive game, but earnestly live it from beginning to end. How could youths better learn to live than by at once trying the experiment of living?"
>
> -Walden, p. 94

> "And you never will, because those things don't exist. No matter how seriously people take them," Thorne said. "A hundred years from now, people will look back at us and laugh. They'll say, 'You know what people used to believe? They believed in photons and electrons. Can you imagine anything so silly?' They'll have a good laugh, because by then there will be newer and better fantasies." Thorne shook his head. "And meanwhile, you feel the way the boat moves? That's the sea. That's real. You smell the salt in the air? You feel the sunlight on your skin? That's all real. You see all of us together? That's real. Life is wonderful. It's a gift to be alive, to see the sun and breathe the air. And there isn't really anything else. Now look at that compass, and tell me where south is. I want to go to Puerto Cortés. It's time for us all to go home."
>
> -The Lost World, p. 416

Eoin O'Hearn
14 December 2025

because of the inextricable relation between observation and quantum mechanics. If an observer is required for all physical phenomena, how can anything else be more fundamental than consciousness itself?

The psychologists have opted for a different track and have attempted to develop ways of analyzing the minds of others. The success of Freud's psychoanalysis dominated the field for the first half of the 20th century, but behaviorism arose to contrast the subjective nature of Freudian nature. Behaviorists attempted to reduce the mind to an input-output machine, with no regards for the internal, subjective experience of the observer. The later fields of functionalism and cognitive neuroscience have filled in some of the holes left by the behaviorists, but many remain.

The biologists have been concerned with methods of detecting consciousness. The growing evidence of agential animals constructing their own niches and experiencing REM sleep indicative of internal dream states, along with many other cognitive functions, caused the field to sign the New York Declaration on Animal Consciousness, confirming consciousness in mammals and birds. In 1973 Dobzhansky published his essay, "Nothing in Biology Makes sense Except in the Light of Evolution" and in a similar way it seems that any explanation for consciousness will have to account for the consciousness found across biodiversity.

As I prepare to graduate in the Spring of 2026, it is reassuring to have found a problem that will accompany me for the rest of my life. I have never been content with our society's push for specialization, and the interdisciplinary nature of the problem will allow me to pursue many different careers and fields of knowledge while still being unified under one umbrella. I know this to be the area I want to continue in because it has felt not that I chose this area, but that I have found it. I wouldn't say that I was aimless, but that I couldn't quite grasp what unified my interests. I explored mathematics, biology, computer science, and many different majors looking for a subject I could latch onto, but it wasn't until I let go of the disciplines that I started to focus on the actual problems at hand. My decision to join the NEW College, which at the time was explained loosely by a desire to take a wide array of classes, has given me the vocabulary to finally talk about the things that interest me as well as the confidence to live at the edge of the unknown, the

1 Acknowledgements

I want to first thank the many professors that influenced and guided either myself or the project over the last year: Dr. Espy-Brown, Dr. Nguyen, Dr. Earley, Dr. Lam, Dr. Tosun, Dr. Galbraith, Dr. Dewar, Dr. Richards, Dr. Alter, Dr. Melkonyan, and Prof. Ford. I am extrememly grateful to them for sharing part of their individual wisdom and specializations.

I would also like to thank the artists whose work appears here: Cade Hopkins, Jack Arnold, Ella Frech, Caroline Gregory, and Corinne Benefield. Their work appears in the following order,

"Me and My World" by Cade Hopkins
"Her Flight" by Jack Arnold
"Return Pattern" by Ella Frech
"Cerberus Fantail" by Caroline Gregory
"Cedar Sunburst" by Corinne Benefield

This work would not be in nearly a similar condition without the feedback from my most trusted editors. Thank you to Diane O'Hearn, Serena Jefferson, and Claire Kennedy.

2 Foreword

My first introduction to these ideas that have garnered my full attention was in PHL212: Ancient Philosophy taught by Seth Bordner. Before I started the class, I held the Ancient Greek philosophers in high regard, though I didn't know anything about them. So, I was delighted to learn that they not only revered my choice of major, mathematics, but also saw them as being more than a description of the world. Plato thought that mathematics was not just a tool for modeling the natural world but was actually the natural world in its purest form. Because these ideas of triangles, shapes, and relationships were everlasting, then they were more real as well—realer than anything that could be perceived in the physical realm. These mathematical forms are a step between the physical reality and the Land of Forms, the truest reality that only contains the perfect and unique Ideals that underpin our reality. Plato's understanding of the universe placed extreme value on the rational disciplines, namely philosophers and mathematicians. Needless to say, I walked out of that class with a big head and newfound respect for Plato, but also a new perspective on my major that planted the seeds for my monomania with consciousness.

If we accept the current scientific understanding of our world and how we arose, the existence of consciousness becomes quite puzzling. At the sub-atomic level, observation collapses the wave equation, energy condenses and localizes to form quarks that themselves make up the fundamental particles - electrons, protons, and neutrons. Atoms collide under the extreme pressure of burning suns, fusing together to produce heavier and heavier elements, as well as photons, heat, and different forms of radiation. Eventually, bonds are produced and molecular compounds float about, but their complexity is restrained by the force of entropy. It is not until self-replicating compounds emerged that complexity is able to be stored and increased over generations of molecular configurations. And finally, the familiar face of evolution appears with the first strands of RNA. Amino acids combine to form proteins and lipids combine to form membranes, all to further RNA's selfish aims. Proto-cells form proto-sensors, eventually able to capture and store the energy of the sun, leading to themselves being captured and stored by other proto-cells.

The first eukaryotic cells emerge as natural factories that minimize energy expenditure and maximize energy obtainment. But selection is cruel and cold, and there is warmth in community. The cells that worked together found and exploited this advantage, developing methods of adhering to each other that would become fundamental for the formation of organic tissue. Tissue became organs that became systems and the complexity of life exploded. Organisms found their niches and specialized, opening the door for other life forms to arise and construct their own niches. Selection drives evolution, forcing organisms to either adapt or die out. Entire ecosystems have risen and fallen during the history of Earth. Extinction acts as a palate cleanser as it wipes the landscapes clean, leaving few organisms. Those left over become the seeds of the next epoch and the cycle renews.

But at what point did consciousness arise? When did cells go from sensing sunlight to feeling it? Why is it that we can introspect on our own experiences? The laws of physics seem to be capable of explaining the entire history of our species, not just as animals in a fight for survival but as a set of extremely complex interactions between self-replicating molecules. But we are left with an intense intuition that we are living, not just surviving. In fact, it seems that consciousness is a prerequisite for understanding our world, including the laws of physics. It seems hard to imagine how we would be where we are without the ability to introspect, learn, and cognitively understand the world around us.

And so, we arrive where humanity has been since we first became aware of our awareness. Different disciplines have created different lines of investigation to answer the different peculiarities of the mind.

The philosophers have largely attempted to solve the puzzling nature of the interaction between the mind and body. It seems that consciousness is of a fundamentally different material than the body, but the two are still able to interact. This curiosity has led them in pursuit of determining the prima materia, the fundamental nature of our reality. Dualists have proposed that everything is either mental or physical, drawing a clean distinction between the two. Materialists have denied the existence of a mental realm, leaning on the success of the laws of physics to explain the world around us. Panpsychists posit that reality is fundamentally consciousness

rest of us pick up the pieces."

"They're not 'playing God', they're using the exact same technology we use to study our fish, just for different results. Or do you have a problem with sequencing our fish as well?"

Bella didn't have a response. Eventually, the server came by and picked up their plates. Louis stood up and looked at Sage.

"Still good to set traps in the morning?"

Sage nodded.

"Good. I'll wake you up. Goodnight, both of you." Louis walked to his room, got ready for bed, and turned his lights out.

3.5 Louis Teaches Sage to Set Traps

Louis woke Sage up, just like he said, and the two walked to the first trap site. Louis carried all of the traps in a large duffel bag on his back, about 120 cafeteria-sized plastic cups with a funnel placed into the opening and holes drilled into the bottom for drainage. He handed Sage a roll of pink reflective tape, grabbed a handful of cups, and dumped the rest on the outskirts of the mangroves. They walked to a clump of mangrove roots and stopped.

"Look for the roots next to the base of the trunk that are just a couple inches off the ground. That's where the fish is likely to be, where it's darkest." He placed a cup into a spot exactly like he described, facing the trap's opening away from the trunk. "You want to use the roots to funnel for you, so the fish runs right into the trap without ever knowing it." He pushed the trap down until the mouth was just under the surface of the water. He placed a couple more traps then asked Sage, "Rip me off a piece of that tape. It's impossible to remember where we put all these traps, so we mark groups of three to make sure we get them all when we come back later." He tied the reflective tape on one of the roots above the traps, and started walking to the next spot.

They continued in this way for a while, but the progress was slow. The mangrove roots crossed over and between themselves, forming knots of tangled wood that

to test positively should he elect to sequence himself anytime in the future. Your new son is legally, and by means of CAS-9 technology, genetically your own. You are completely responsible for his well-being until his eighteenth birthday, at which point you may decide whether you wish to continue providing any parentage. Do you understand?"

The couple was too enamored by the small fingers already beginning to clasp around the thumb of Mrs. Pith to listen any more attentively. They had heard this speech dozens of times before in front of CAS-9's army of lawyers, and even once from the CEO herself. They nodded and signed the documents Mr. Solon slid in front of them. Before their signatures had dried, the lawyer packed them back into his suitcase and left without saying goodbye. The Pith's didn't notice.

3.2 The Crew Arrives on The Island

"We're coming up on it now!" Louis pointed to the island rapidly nearing them.

It was small, small enough to kayak around, and Louis said only a few other people lived there besides the small diving companies. Sage took a deep breath of salty air. She could feel the warmth leaving her skin as the evening faded and the rock of the boat waned as they closed in on the dock. Pelicans and Frigate birds circled overhead, diving occasionally and without much success. Just beyond the first of the palm trees, she could see the front of the building. The building further defined itself as a large eating area with rooms off to the side. It was obvious many repairs had been made recently. The fresh, tan wooden planks contrasted with the darker planks that made up most of the building. Sage wondered how long the weathering process took. Louis had mentioned that a hurricane had hit the island since he was here last, but she wasn't sure how long ago that had been.

Once the boat had been tied properly, they began unloading the crew's baggage. They each had a personal item, but the majority of their bags contained equipment they would need for their short time on the island. The others had packed most of it, but Sage had seen stereoscopes, trap cups, and a wide array of colorless liquids. She had read the project proposals but was inexperienced compared to her coworkers.

Louis was her research advisor. He specialized in animal behavior, using the mangrove rivulus fish as a model organism to test hypotheses. He originally began in evolutionary game theory, which is why Sage had decided to join his lab as opposed to the many other labs wanting undergraduate help.

Bella was a PhD student in Louis' lab. She was finishing up her thesis and needed a field work component, which sparked the formation of the trip they were currently on. Sage hadn't spent much time with Bella previously because they worked on separate projects in the lab. They met each other for the first time packing cup traps and seemed to get along fine. With the luggage unloaded, they grabbed what they could and headed to their rooms to unpack.

3.3 Louis Shows Sage and Bella the Island

Sage dropped her bags off in the room and then left to get a look at the island. The sand was soft but littered with bits of plastic that had drifted there. She leaned down and grabbed a handful, letting it filter slowly through her hands. She dropped the pieces left in her hand and felt the temperature of the water. Cool, but not freezing. Perfect for a refreshing dip after the work to come in the humid and muggy mangrove habitat where they would, presumably, find the same fish Sage knew from hours in the lab.

She had joined the lab as a second-year undergraduate. Like many, she was interested in grad school but wasn't sure if research was right for her. This trip was a personal test she was running. Would she fall in love and commit to a life of academia? Or would she find another dead end in her search for a career? Her pondering was interrupted by Louis.

"How's the water?"

She turned to look at him. He had changed his clothes into just a swimsuit, and he saw her reaction.

"Why would I come to an island and not immediately jump in? Have a bit of respect for the beauty of this place." He laughed and ran into the water.

Sage watched him splash around for a second before Bella came to join her. "Is

parents were seen as best of the best, as people that had gained every certification, licensure, and diploma they could get their hands on to prove they were qualified to foster.

Suddenly, it seemed like everyone was considering fostering. The rising cost of living made it too expensive to raise your own, but there were still millions of young adults wishing to start a family. Fostering just made sense. The government practically paid for anything the kid could want, within reason, and many were happy to have the weight of feeding another mouth taken off their shoulder. Young couples didn't have to make the tough decision between aborting or changing their life goals anymore. There was always another family looking to take in the newborn.

But as more and more chose to foster, the birthrate plummeted. The demand for newborns far exceeded the supply and the waitlist started to grow. The average age of new parents slowly started to increase until no one under thirty-five was enrolling their kid into the first day of school because anyone younger couldn't afford to have their own child.

To combat this, the government released its grip on the genetic industry. Companies had long wanted access to the undifferentiated cells overflowing in every infant, but had been contained thus far. Fears regarding genetic privacy and its potential abuse were replaced by fears of a declining population in the working class. Now, however, companies like CAS-9 flooded the market with deals promising to pay for the costs of raising a child, just as long as it wasn't the one you gave birth to. Anyone looking to start a family could make a deal to exchange their newborn for another, and the company would pick up the bill. Slight genetic modifications would be made so that there would be no physical differences, and even genetic sequencing could be fooled.

"Who cares?" A wave of disgust broke across Bella's face. "I'm sure all those kids growing up without their parents care."

"They are growing up with their parents though. The swap takes place before the first memories are even made. Both families go home happy and the company gets to make a couple extra bucks."

"But it's wrong! These companies can't just go around playing God while the

come up with this?"

Louis answered, "We didn't, at least not originally. It was done first with a species of mongoose a decade or so ago. We just wanted to try it with our fish because of its ability to clone itself. It will help us get a grasp of whether or not the prosocial behavior is just a result of genetics and is a part of inclusive fitness, or if it's something else entirely." With that, Louis and Bella shared a knowing glance, but Sage didn't quite pick up on its meaning. She was still trying to come to terms with the idea of parent swapping. "You following, Sage?"

"Yeah...I am.. it's just that... it's a bit morbid, no? What with separating the fish with their kids and stuff. I know their fish but..."

Louis shrugged, "it's really not that different from what's going on right now in the real world. You guys heard about that family, didn't you? The ones that traded their child? The..." Louis trailed off while trying to remember the family's name.

"No one has done that," said Bella. "And even if they had, it's not fair to compare that to what we are doing in the lab." Sage gave her a questioning look. "We aren't animals. The fish we study don't care if their mom is their mom or their dad is their dad, they just need someone for those first critical periods of their life and then, bam! They're off doing their own thing. The fact that those people traded away their own child..."

"The Piths!" Louis blurted, apparently having figured out the name that escaped him.

"What?"

"The Piths! That's the name of the family. And Bella, it is the exact same with what we're doing. Why else would those companies offer to pay you to raise a child? They want you to trade your kid for another so that they can get a sample of those stem cells while they're passing those kids around. Yeah, nobody knows what they're doing with 'em, but who cares? You can't tell the difference anyway."

It was true. In the last couple decades, billions had been poured into updating the foster system. It started first by raising the amount they sent to the families per child, but then they had to improve the vetting process because the program was getting to be too popular. With those improvements, it wasn't long before foster

he always like this?"

Bella laughed. "Yes, but you get used to it. And for all his quirks, his work is really good. You heard what he was saying about the sites around here? He's been here more times than you would expect, he's done this for a long time."

Louis walked out of the water, dried off, and put his shirt back on. "Ready to go explore the island? I'll show you where we'll set traps tomorrow." The two women nodded and followed him behind the rooms where they were staying.

Although the island was small, it was densely covered by mangroves and other tropical flora. While this was great for the rivulus, it was not so great for navigating around. Thin paths had been cut, just wide enough for two people to walk side-by-side. Louis was walking in front while Sage and Bella walked together. The paths consisted of one main loop around the island with offshoots giving access to the few houses on the island, as well as the sites where they would be collecting their fish. They were careful to remember the names Louis called out as they passed each site, but they all looked similar. Dense mangrove roots sitting just above an inch of standing water. Perfect for their fish, and mosquitoes. Thankfully, the insects weren't bad and largely remained within the dense trees, not daring to leave them for more than a foot.

They could see no movement, but that didn't mean much. The fish they were looking for was a scientific wonder. Not only was it born with both sets of reproductive organs, meaning it could asexually reproduce, but it seemed to not even be a fish at all. One researcher reported specimens living for months out of water, surviving in the moisture of a rotting log. There was a reason they were choosing to trap rather than any other means of collecting, the fish were just plain hard to find.

They passed the last site and continued along the loop to head towards their rooms. The path had opened up onto the shore, and the crew had taken off their shoes to enjoy the ocean waves. Ahead of them was one of the few houses on the island, and the owner was outside enjoying the water as well.

"Hello" greeted Louis when they got near.

The man turned. "Hello, who are you?"

"We're biologists staying at the resort just past your house. My name is Louis."

He stepped forward to shake his hand. The man looked at him for a second, then shook his hand, "Mine's Phillip. Biologists?" Phillip lowered his hand and looked at Sage and Bella.

"Yep, biologists. We're studying prosociality and blind paternity using the mangrove rivulus fish." Bella smiled as she said this, proud of the work that would become her dissertation. Phillip stared. He was not quite as impressed. Bella, slightly unsettled, continued, "Do you live here?" Phillip silently nodded, then slapped a mosquito against his arm. "Oh, I bet you get those a lot. They must get annoying," laughed Bella, hoping the stranger would sense her unease.

Phillip shrugged. "They're a part of life, just like any other creature. Should't a team of biologists know that?"

The three looked at each other, not knowing how to respond. Then, Louis said "Well, it was nice to meet you, but we'd better keep heading back to make it for dinner. Have a good one." With that diversion? the crew walked briskly past him and towards their rooms.

Sage looked back before they entered the mangrove paths and saw Phillip standing still right where they had been talking, unmoved since they left. "Man what the hell is up with that guy?" she asked.

Louis replied "Yeah I don't know. I don't remember him living here the last time but wow, he gives me the heebie jeebies." Bella and Sage murmured in agreement. They were all ready to get back to the resort and forget about the strange interaction.

3.4 Bella and Louis Talk Methodology

When the crew finished finding their seats around the table, the server brought the soup out. Included with the stay were three daily meals, always typical Belizean meals. Louis picked up his spoon and started to drink the broth of the soup before grabbing his fork and eating the fried fish sitting in the middle. Sage and Bella hesitated before finally being brave enough to take the first bite. Though salty, the soup was amazing and all three ate a large portion of their meals before beginning to talk.

"So, how'd you two like the mangroves?" Louis asked while resting his spoon. The two took a second to respond, but then Sage said, "I absolutely loved it. Such a crazy environment..." trailing off as she looked for the words to explain it.

Bella replied similarly, and Louis started to explain what the following days would look like. "We'll have to go out every morning to set traps during the cool of the day and then collect them before they've spent too much time baking. Of course, if we find all our males and hermaphrodites then we can relax on trapping and focus on packaging, but I doubt that will happen."

Bella asked "You don't think we'll get enough fish?"

"It's not that, I don't think we'll get enough males. Rivulus don't like to maintain a significant percentage of males, most prefer to be hermaphrodites. These populations have some of the highest rates of males, compared to populations in Florida, but even still only get up to about 25

"I don't," Sage said.

"That's fine, we can go out tomorrow morning while Bella sleeps and I'll show you."

"That sounds good, but what happens after we get the fish?"

"We'll package them up so they can travel safe back to the lab and then start the behavioral assays. Bella, you wanna explain this part?"

Bella nodded, caught in the middle of a spoonful of broth. She swallowed, then said "We are testing the effect that blind paternity will have on colony weight distribution. Usually these fish hatch and raise their own spawn, like most species. Instead we'll swap eggs and parents around so that none of the fish can tell whose kids are whose. Then we will provision half of the parents with food and half not, so that half of the population is given a sort of head start. After the kids reach maturity, we'll measure body masses and compare the distributions." Sage looked confused, but Bella just continued,

"The idea is that if we see a uniform distribution—everyone is roughly the same weight— then, for one reason or another the parents that got extra food at the beginning decided to help out the less fortunate."

Sage slowly began to wrap her mind around the experiment. "How did you

3.8 Louis Arrives at Home

Louis dumped his suitcase on the ground and collapsed onto the bed so his dogs could give him the welcome they were waiting for. After the wagging slowed down, he sat up to unpack, knowing he would never do it if he didn't just get it over with. In the rush of packing everything to get onto the only boat leaving that day, he had thrown everything in without folding or organizing anything, and was paying the price. He threw all of his clothes into the washer, and started on the equipment bag. His waterproof book, computer, pens, and all other research materials found their place on his desk. At the bottom of the bag, he noticed some empty vials that he packed last second before heading to the island, just in case they had better retrieval rates then he had expected. He pulled them out, along with the receipt from the company that had funded their research.

THERMO SCIENTIFIC Inc.

Item	Quantity	Price	Cost
4" falcon tubes	12	24.99	299.88
Stereoscope (20x)	3	195.00	585.00
Paraffin wax (block)	10	6.99	69.99
DMSO (1L)	2	30.00	60.00
Billing Address	CAS-9 19505 Biscayne Blvd Aventura, FL 33180	Total	1,014.87

made it hard to walk across. The soil beneath was no better. The massive knots slowed the incoming tides so much that the water was essentially stagnant. Over time the water seeped downwards, liquefying the soil. In some places, the water had been undisturbed for so long that the ground was meters below the surface. But, it was impossible to judge how far you would sink just from its appearance, meaning that the two had to trapeze over the roots rather than risk sinking into their hips. Louis said one time he had sunk to his waist, becoming completely trapped while the tide rose around him. Sage shuddered to think about it, but that wasn't the only thing weighing on her mind.

"Do you think we'll get enough males?" Sage asked as she handed Louis another piece of tape.

Louis tied it off and shrugged, "I'm not sure. It's always a guess coming out here. Some years are better than others, but I haven't been here for a while."

Sage thought for a minute then asked "Why aren't there more males? Shouldn't it be fifty-fifty?"

"Why do you think?" He always liked the chance to make his students think things out.

"Is it because they aren't born male, and so only some of them actually transition?"

"Kinda. There just isn't a need for so many males. The hermaphrodites can take care of most of the birth requirements needed to keep the colony going. It's only every once in a while that the genes need a recombination and thus, you keep a couple males around.

"You've got to remember that it's not just individuals that face natural selection. Everything in Nature has to prove it belongs there, even things we might not think of at first. Consider humans and our 50/50 split. There's about as many human males as human females, but this isn't by accident. It's the best strategy for survival. Too many males and the population can't reproduce fast enough to maintain itself. Too many females and there's not enough variation in the male's genetics to fight inbreeding depression and the accumulation of deleterious alleles. It's a balance, just like every other aspect of an organism, of a characteristic of the population

itself, but that doesn't mean evolution is somehow blind to it."

They continued to set traps, moving slowly across the mangrove roots. Sage handed Louis cups and tape and he set the traps. Gathering her courage, Sage asked "Do you really think that kid will be alright?"

"Who? The Pith's new kid? Yeah, why wouldn't he be?"

"I don't know, there's just something freaky about swapping your kid."

Louis laughed, "Yeah it's new to me too, but that doesn't mean it's wrong. Hell, if I did, I wouldn't be doing it to these fish."

"But don't you think it's different? For humans, I mean."

"No. And why would it be? Humans are part of the animal kingdom just like fish and just like every other living thing that walks on Earth. The fact that we can talk about it doesn't make any difference. Our behavior is determined by the same laws as everything else."

"What do you mean?"

"You know how some people are night owls and some are morning people?" Sage nodded. "Which are you?"

Sage thought about all the nights she had stayed up late going out or talking with her friends, just to wake up groggy for her biology class the next day. "Umm... a night owl, I guess" she admitted sheepishly.

Louis laughed. "Me too. And so are a lot of other people. But there are also lots of people that like getting up early. Those are different chronotypes, different circadian rhythms, and they're determined genetically. Get one set of genes and your body produces the hormones to wake up and go to sleep early. Get another, and your hormones spike at night instead.

"Now think about all the different things you associate with morning people, cheery, proactive, disciplined, maybe uptight even, vs. the things you associate with night owls, creative, independent, maybe lazy.... the list goes on. Whole parts of your personality are determined by a small set of genes that tell a specific hormone to fire at a specific time. Extend that thinking across the thousands of genes that a human has and it appears your personality might not be so personal, but just a result of the genes you were given at birth. As immutable as your skin or your hair type."

3.7 The Crew Leaves The Island

The crew continued like this for the rest of their stay. Louis and Sage set traps in the morning and then Sage returned with Bella to collect whatever they had managed to trap. The male-to-hermaphrodite ratio ended up working out in their favor; Bella had enough males to run enough trials so that her statistical test would have the power she needed to publish in a good journal. Sage got used to moving eggs between different clutches, and once the fish calmed down, they didn't seem to mind either. They hadn't been back to the mangroves after collecting all the males, so the mosquito bites came and went. In fact, it wasn't until they were packing their bags again, getting ready to return home, that she even thought about those insects. She packed the repellent into her bag, grabbed the remaining bags full of trap cups and stereoscopes and walked out to meet the rest of the crew already on the boat.

Sage looked back at the island. She thought about all the things they had talked about while collecting fish. Where do humans fit in the vast diversity we understand Nature to be? Was Louis right? That we are nothing more than interacting molecules that make us up? Or was Bella? That rationality and pure reason are gifts to be used to order the world around us? She didn't know the answer, but she was glad that she knew the questions.

The captain unhooked the mooring rope and started the motor. As the island started to leave them, she saw Phillip standing in front of his house, just where they had left them. He turned to meet her eyes, and she could see the swarm of mosquitoes approach him from behind. They covered his skin until she could see nothing but his eyes. She shuddered to think about the bites he must be suffering, and jumped when she felt just one land on her arm. She slapped it, but was slow to move her hand. She thought about what he had said, about even mosquitoes just being one part of life, neither good nor bad. Louis and Bella would continue to argue their theories until the end of time, but Sage was happy to stay in the real world, to smell the salt in the air and feel the sunlight on her skin. She closed her eyes and let the rock of the waves drift her off to sleep.

at the entrance to the mangroves. She wanted to go cool off like Louis had done when they first got to the island. She zipped the bag closed and started to make her way over the roots.

The mosquitos appeared before Sage had a chance to prepare. In seconds, twenty, then thirty, then upwards of forty bloodsuckers were hovering around her, choosing the best place to attack. The first of them landed on the bare skin of her arms, but the light clothing needed for research in the muggy mangroves meant that her entire body was fair game. They covered her legs, hands, feet, back, and eventually were courageous enough to land on her face.

Sage felt a sense of panic rising as hormones flooded her system in response to the constant, deafening sound of their buzzing. The number of mosquitos continued to grow, and soon she struggled to see her own skin beneath the forming layer of bugs. Her heartbeat and pace both sped up, fighting to get her out of the wooded corridors. She swung her arms as she ran, trying to swat at all parts of her body, but it wasn't enough. As soon as she cleared the back of her arm, they began biting her calf. They landed on the empty space faster than she could clear them, ready to drain her milliliter by milliliter.

The fight turned mental. Sage struggled to keep her focus. Every bite sent emergency signals to her brain, derailing any train of thought she could begin to muster. Adrenaline production rates continued to increase. Dozens of bumps began to swell. She ran faster, but the mosquitos were everywhere, buzzing and biting, threatening to send her into complete panic. The body took complete control from her mind, relying on instinct alone to save her from being exsanguinated. They became braver and braver, and the ones landing on her face sent shocks to her system. Primeval emotions welled up from the deepest recesses of her brain.

They left when she made it to the shore. The last stragglers stopped before flying over the water. Sage submerged herself, hiding in the cool ocean from the airborne pests.

Sage was silent, contemplating the implications of what Louis was saying. Could it be true? Could her likes, dislikes, attitude, even everything, be just a part of her genes? No, it couldn't be. "Well what about the effect of the environment? I mean, it matters whether you grow up rich or poor, or in one country or another."

Louis smiled. "Yes, but that's exactly the point that I'm making! When we design an experiment for our fish, we consider the genetic and the environmental effects. Why should it be any different for humans? They thought about all that before giving the Pith's their new kid. That kid gets a set of parents and the parents get the kid they've always wanted. At the end of the day, their family is just as happy as one made the natural way. We humans have a chip on our shoulder about being above the natural world, but we're a part of it whether we like it or not.

"Before evolution was discovered by Darwin, people used to believe that all life on earth could be ordered from less to more intelligent, with humans being one step below god and one step above all other animals. But the discovery of evolution being driven by natural selection forced us to recognize our own reflection in Nature. And yet we still grasp at straws, trying to claim that for some reason we are special, that somehow, of all others, Homo sapiens is divine."

Louis placed the last cup and the two headed back towards where they came. Sage kept quiet, thinking about all he had said, about humans refusing to accept their organic origins. They grabbed the empty duffel bag and headed back towards the rooms, but they didn't make it far before they were reminded of the insects around them.

Instead of retreating back into the mangroves after they went farther down the path, the mosquitoes stayed. They tried to swat them as they walked, but the buzzing cloud refused to dissipate. Only a few dared to leave the bunch and make an attempt at biting them. Sage thought she had gotten them all before they managed to make contact but saw spots rising on both her arms. She didn't like the thought of them being able to escape her attempts, silently sucking her blood. When they got back to the room, Bella came out of her room and met them for breakfast. They ate quietly, then made plans for Bella and Sage to go out that evening to collect the traps they had just set.

3.6 Bella and Sage Collect Fish

"Ready to go?" Bella looked at Sage, trying to gauge her tiredness. Louis had stayed behind to nap, but Sage insisted on going.

"I'm just getting warmed up. Come on, I'll show you where we set those traps." The two left Louis in his hammock and walked towards the site they had been that morning. Sage hadn't talked to Bella since she had that argument with Louis over the Pith family. They both felt it sitting in the air between them, adding to the effect of the humidity. Finally, Bella spoke first.

"So, did you guys talk anymore about that kid?"

Sage wasn't sure if Bella was testing her or wanting to clear the air, "Um, yeah, for a bit. But not really anything new."

Bella nodded, then asked "Did he give you his night owl argument?"

Sage was shocked. "How'd you know that?"

"He gives that same argument anytime anyone brings up humans in regards to evolution or whatever. I heard it for the first time when Louis was arguing with a postdoc about whether or not altruism really exists." They both felt the tension leaving the air. They made it to the trap site and began collecting cups. Bella walked through the mangroves calling out the amount in each trap while Sage wrote everything down and put the cups back in the duffel. It went faster than setting, but was still slow. This time Sage didn't care. She was glad for the chance to keep talking to Bella. She decided to risk opening the topic again.

"So you think Louis is wrong about the night owl stuff?"

"No, not that exactly. I've read the papers, and the different circadian rhythms thing is true, and it is genetic. I just think Louis takes it too far. He's convinced that our whole personalities can be studied like we study the behavior of these animals. Mess with a hormone here or a receptor there and see the effect it has." "Oh, okay. So you think we're more complicated?"

"Totally. Yeah, sure, we are a part of nature and all that stuff he likes to say, but to reduce us to nothing more than Homo sapiens is ignorant of our civilizations today. We've completely removed ourselves from the forces of Nature. Food, water,

shelter, all those things we need to survive? They're a given nowadays, at least for most parts of the world. Survival of the fittest doesn't mean much to us anymore, practically everyone is the fittest. Just look at our fossil record. For hundreds of thousands of years our skeletal system, our physical makeup, has been pretty much the same. Our tools are the things that are evolving now, now us.

"And that's for good reason. No other species on Earth has both the cerebral power and dexterity to conquer the natural world like we have, otherwise they would have. We are alone on this planet for all intents and purposes. Even our closest genetic relatives, the bonobos, are millennia behind us in terms of civilized society, if they'll ever get there."

"But then why do you have a problem with what the Piths are doing? I mean, if we're so advanced, can't we just kinda do what we want?" asked Sage.

"Sure we could, but why? We don't have to think about fighting for survival like everything else does, but that's only because our brains have evolved to the point where we can think rationally rather than strategically. It's why we have the technology that we have. We rely on the pure truths mathematics and physics describe to separate us further and further from the chaos that is the natural world. The scientific method has cured diseases, built, then destroyed, civilizations, and gives us a reason to be out here. Louis thinks that our personalities are tied to us genetically. That the morals, ethics, and ruling ideals of society are nothing more than the complex interaction of hormone transport chains and environmental conditions. But it's our pursuit of what is rational, what is everlasting, what is unnatural, that has led us here."

Sage put the last of the traps into the duffel bag as she tried to understand Bella's ideas. She had been convinced by Louis' argument, but doubts were starting to creep in. Science has always been the way she interacted with the world. Since she was young she'd always excelled in those classes, internalizing the importance of objectivity and fact. But is objectivity possible? If all experiments are observed by a person whose thinking could be affected by their genetic makeup, doesn't that make bias inherent? And if that bias was inherited, whose to say we aren't blind to it as a species? Her head was swimming, and thankfully Bella was waiting for her

Theorem 2 Let $r = 1$ and C, D be disjoint subsets of V. Then $P_C + P_D = P_{C \cup D}$

Theorem 3 If $r = 1$, then $\rho = \frac{1}{N}$ for all graph structures.

Theorem 4 Let $C \subseteq V$ and $r > 1$. Then

$$P_C^{(1)} \leq P_C^{(r)}$$

While determining the fixation proability of a given mutant has been a central problem in Evolutionary Graph Theory, determining the time to fixation has also been an area of focus. The first main result determined the fixation time for isothermal graphs, the doubly stochastic cases where the sum of the incoming edges is equal to the sum of the outgoing edges at each node.

Theorem 5 Let G be an isothermal evolutionary graph with large N. Then the meant time to fixation for a mutant with fitness r is

$$t_1 = \sum_{n=1}^{N-1} \frac{(r^N - 1)(r^{N-n} - 1)(1 + r)}{(r^N - 1)(r - 1)}$$

Although it seems to intuitively be true, there still lacks a formal proof for the following conjecture regarding the relationship between the mutant fitness and fixation probability.

Conjecture 1 The fixation probability, ρ, is monotonic with respect to the mutant's fitness, r.

4 Review of Evolutionary Graph Theory and Games on Graphs

4.1 Introduction

The field of Evolutionary Graph Theory first arose with the seminal paper in 2005 titled "Evolutionary Dynamics on Graphs" by Lieberman et al. [Lieberman et al., 2005]. The authors seeked to understand how population structure would affect the evolution of newly arisen traits, either supporting their fixation or helping to drive them back to extinction. To study this behavior, homeogenous populations are generated with a neutral fitness equal to 1. Then an invader, or mutant, with fitness r is randomly chosen to replace one of the individuals. Two nodes are then selected, one to reproduce and one to die, the graph is updated, and the process continues in this way until the mutants go extinct or reach fixation. See Figure 2 for an overview. The process is built upon an underlying stochastic structure that first appeared in Moran (1958) [Moran, 1958]. The field has seen a continued surge of interest and new areas of study have explored in determining the computuational complexity of fixation probability problems, comparing update rules, calculating mean fixation time, and game theoretic extensions.

Game theory can be included with Evolutionary Graph theory by having nodes face strategic interactions with the neighbors. The payoff of the game then affects the nodes fitness and so strategy profiles can be dynamically evaluated on the graph's structure. Studying evolutionary stable strategies is a central problem in evolutionary biology. Organisms, in their fight for survival, can be viewed as playing different strategies to maximize their chances at aquiring resources or minimizing energy expendtiture. A classic example of this is the 'Rock-Paper-Scissors' game represented by the sexual strategies of the male Side-blotched lizards (Uta spp.) [Sinervo and Lively, 1996]. In this lizard, there exists three different throat-colors, orange-throated, yellow-throated, and blue-throated. Each color attempts to mate and reproduce via specific strategies. The orange-throated males are aggressive and establish large territories. Yellow-throated males are also known as sneaker

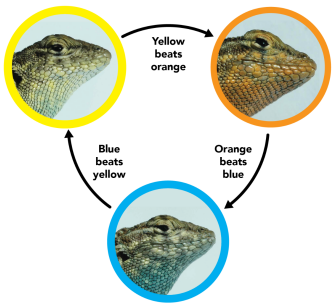


Figure 1: Dynamic relationships between different color morphs. Reproduced from [High-stakes, 2026]

males because of their tendency to sneak into the territiories of the orange-throated males and mate with the females. The blue-throated males defend just one or two female mates, but are able to defend against sneaker attacks.

These strategies are all face frequency-dependent selection. It is more advantageous to be a rarer morph than to be playing the same strategy of the other male lizards. This type of selection has led to a six-year period in which the percentage of different color morphs in the population oscillate around each other, hence the reference to rock-paper-scissors: orange beats blue which beats yellow which, in turn beats orange, and so no single strategy can reach fixation in the population.

In this case, the forces driving the favorable selection is apparent, but for many other extant strategies this is not true. Classical evolutionary theory seems to not allow for the selection of non-selfish behavior. In 1964, W.D. Hamilton introduced the idea of 'inclusive fitness' [Hamilton, 1964]. This led to the development of "Hamilton's rule" that defined mathematically the situations in which cooperative behavior would be favored by natural selection. Let r be the relatedness between two individuals, b the benefit of cooperative behavior, and c the cost to the helper. If the

ness is not an easy thing to do. In fact, for most graphs the calculation complexity is NP hard, and so we turn to imposing restrictions on the parameters of the graph. [Lieberman et al., 2005] found that for a special subset of graphs, named isothermal, the fixation probability relies solely on the relative fitness of the mutant and not the underlying structure of the graph. We say that a graph is isothermal if the sum of the weights of the edges leaving a vertex is equal to the sum of the weights of the edges entering a vertex, hence isothermal. This gives our first theorem.

Definition 3 An evolutionary graph G is isothermal iff

$$\sum_j w_{ij} = \sum_j w_{ji} \quad \forall\, v_i$$

Theorem 1 An EG is isothermal iff the fixation probability of a randomly placed mutant is ρ_1.

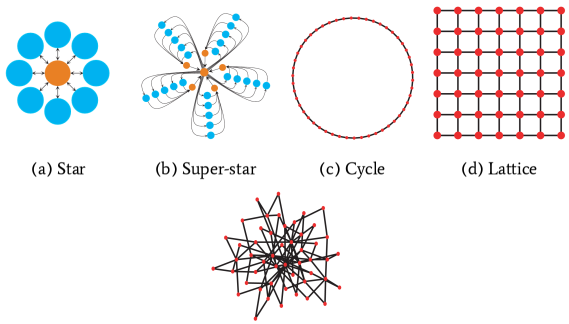
Seeing as this is still a small subset of the total possible graph structures, we turn to alternatives to determing the fixation probability, ρ. As mentioned before, we can consider the case when more than one individual mutates and so a subset of the population has fitness not equal to one.

Definition 4 Let $C \subseteq V$ and $r > 0$. Then $P_C^{(r)}$ is the fixation probability given that all individuals in C have been replaced with mutants with fitness r.

Using this definition, we can relate the fixation probability of a subset of a graph to the general fixation probability, ρ, using the following equation.

$$N \cdot \rho = \sum_i P_{v_i} \tag{4}$$

Going futher, it was found that the subset fixation probabilities are additive under neutral drift, when $r = 1$. From Equation 4 and Theorem 2, the fixation probability under neutral drift can be shown to always be $\frac{1}{N}$. Additionally, the fixation probability under neutral drift can be used as a lower bound for determining the fixation probability of an advantageous mutant, $r = 1$. Theorem 4 states this formally.



(a) Star (b) Super-star (c) Cycle (d) Lattice

(e) Scale-free

Figure 3: Different Graph Structures. Graphs (a) and (b) are reproduced from [Lieberman et al., 2005]. Graphs (c)-(e) are reproduced from [Ohtsuki et al., 2006].

A single node has been chosen to be the invader, but in general not all nodes are equal. The degree of the intial node can have an impact on the fixation proability of the mutant, regardless of its fitness. So first, consider a graph where are all nodes are equal. In a well-mixed population each vertex is connected to every other vertex, and so all have equal degree, $N-1$. The simplest case, of an undirected graph, gives that $w_{ij} = w_{ji} = 1$ for all i, j. This special case is the Moran Process of [Moran, 1958] and gives the Moran probability, ρ_1, that a mutant with relative fitness r will reach fixation. This probability has been determined to be

$$\rho_1 = \frac{1 - 1/r}{1 - 1/r^N} \tag{3}$$

But what of populations that are not well-mixed? Such is the case of most graphs, and much of the work in Evolutionary Graph Theory has been focused on computing the fixation probability of graphs with differing structures. The most commonly studied graphs are the star, super-star, cycle, lattice, and scale-free graphs, displayed in Figure 3.

In general, calculating the fixation probability of a mutant given its relative fit-

following relationship holds, then organisms will act altruistically,

$$rb > c \tag{1}$$

We can extend this idea to general populations of organisms by playing the "Prisoner's Dilemma" game on an Evolutionary Graph. In the famous game theory problem, individuals choose to either cooperate or defect. Their playoff is determined by both the strategy they choose and the strategy chosen by their opponent. Cooperators choose to help and incur a cost in doing so, while defectors solely benefit. Ohtsuki et al. found that cooperation was favored by selection when the benefit-to-cost was higher than the average degree of the graph, k [Ohtsuki et al., 2006].

$$\frac{b}{c} > k \tag{2}$$

This relationship was shown to exist in population structure of primates and so may have contributed to the selection of cooperative behavior. Voelkl and Kasper (2009) analyzed the interaction dynamics of 70 different primate groups [Voelkl and Kasper,]. They found that the resulting graph followed the benefit-to-cost ratio rule from [Ohtsuki et al., 2006].

This paper reviews the main findings of Evolutionary Graph Theory and formalizes the inclusion of game theoretic strategic interactions. It finishes with a discussion on numerical simulations and evidence for benefit-to-cost ratio rule being necessary, but not sufficient, for the fixation of cooperation in a population.

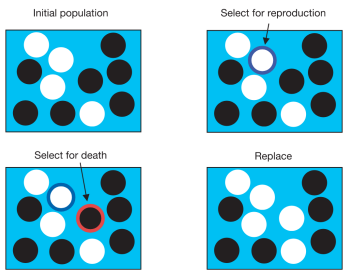


Figure 2: Updating an Evolutionary Graph. Reproduced from [Lieberman et al., 2005]

4.2 Evolutionary Graph Theory

Definition 1 A population V is a set of N individuals,$\{v_1, \ldots, v_N\}$, along with a fitness mapping $f : V \to \mathbb{R}$ such that $f_i = f(v_i)$ is the fitness of individual i. We say that an individual is a resident if $f_i = 1$ and a mutant otherwise, $f_i \neq 1$.

Definition 2 An evolutionary graph G is a population V with adjacency matrix W such that $\sum_j w_{ij} = 1 \; \forall \; i$.

As the reader might be able to guess based on the stipulation on the sum of the rows, we can consider evolutionary graphs to be a stochastic process. At each time step, an update rule is used to determine which vertices, or individuals, are chosen for birth and death. The simplest such update rule, Birth-Death with Birth bias (BD-B), is introduced and used in [Lieberman et al., 2005]. In this update rule, one first picks vertex v_i with probability proportional to f_i. Then vertex v_j is chosen from the neighbors of v_i with probability w_{ij} from the adjacency matrix. Lastly, v_i replaces v_j so that v_j is now identical to v_i, but still a separate node, or individual.

This is, however, just one such update rule that can be used when studying evolutionary graphs. There exists three main families of update rules. BD-B is part of the Invasion Process family (IV), where the first node chosen replaces the second. The reverse, where the second node chosen replaces the first and so v_j would replace

Table 1: Possible biases with Birth-death update rule.

Bias	Process
Unbiased and undirected	Pick v_i and v_j with uniform probability
Directed	Pick vertex v_j with probability proportional to outgoing edges of v_i
Birth-bias	Pick vertex v_i probability proportional to f_i
Death-bias	Pick vertex v_j probability proportional to f_i^{-1}

v_i, constitutes the aptly named Voter model family (VM). The last family focuses on choosing edges rather than nodes, and so is named Link Dynamics (LD). In this family of update rules and edge is first selected between nodes v_i and v_j. Then, one node is chosen to replace the other depending on what bias rule is being used.

Reviewing the first update rule, BD-B, the first node is chosen with probability proportional to its fitness, but the fitness of the node chosen to 'die' is not considered. There exists a fitness bias on the node chosen to reproduce, hence a Birth bias. But we could easily choose to instead consider put the bias on the node chosen to die using a probability distribution inversely proportional to its fitness, where a low fitness would increase the probability of being chosen for death.

The possible biases are further outlined in Table 1 for the Birth-death update rule, but are analagous for the other families, with one exception. When using a Link Dynamic update rule, it is possible to have an Imitation bias such that whichever node of the selected edge has the higher fitness is chosen to reproduce.

Now that we have defined the components of an evolutionary graph and have defined an update rule, we can now begin to consider questions such as "With what probability will a mutant with relative fitness r fixate in a population of N individuals?" We first consider the case where a single individual v_m is chosen uniformly from the population V to have fitness $f_k = r$. Later on, we will see extensions where more than one individual is chosen to mutate. In those cases, the specific subset that is chosen to be replaced can have a large effect on the fixation probability, but for now we return to the case of a single invader.

4.3 Games Theoretic Extensions

Continuing with [Shakarian et al., 2012], we can begin to consider game theoretic extensions of evolutionary graph theory. When considering evolutionary stable strategies in standard evolutionary game theory, the populations are often considered to be well-mixed. We see, however, that considering other population structures leads to different stability results.

Evolutionary stability is a primary focus of evolutionary game theory. A strategy is said to be evolutionary stable if it satisfies two conditions. (1) a population of organisms playing that strategy is resistant to invasion by another strategy and (2) that the strategy is favored by natural selection. (2) is equivalent to saying that the fixation probability, ρ, is greater than that of neutral drift, i.e. $\rho > \frac{1}{N}$.

When beginning to examing games on graphs, we first must make a relation between the payoff of the game (P) and the resulting fitness of the organism (f_i). This is done with the following equation with parameter α. This parameter determines how much influence the payoff of the game has on the fitness of the individual.

$$f_i = 1 - \alpha + \alpha \cdot P$$

If $\alpha = 1$ then the fitness of the organism is equal to the payoff recieved from playing the game. If, however, $\alpha = 0$, then the payoff has no effect on fitness. Most applications consider cases of weak selection, i.e. where $\alpha << 1$, so that the game has a minimal, but still apparent, impact on the fitness of the individuals.

We begin like all treaties of game theory, with a discussion on the prisoner's dilemma, shown in its normal form below. Two players are offered the choice, simultaneously and independently, to either defect or cooperate. If they both cooperate, they each earn payoffs of 2. If they both defect, they earn a lesser payoff of 1 each. If player 1 chooses to cooperate and player 2 chooses to defect, then player 2 walks away with 3 and player 1 walks away with nothing. Similarly if player 1 chooses to defect and player 2 chooses to cooperate then the first player will walk away with 3 and the second with none.

	cooperate	defect
cooperate	2,2	0,3
defect	3,0	1,1

This game has been studied extensively because of its far-reaching implications in many different fields, especially evolutionary biology. Many species exhibit, and often rely upon, cooperation in their social interactions, but how could this have arisen within Darwin's natural selection framework? Why would an indiviudal choose to cooperate and potentially walk away with nothing? It seems that defectors have the stronger strategy as they can always ensure they walk away with a positive payoff.

In fact, in standard game theory this is the conclusion we reach. Both players are playing their best strategy when they are defecting, corresponding with the formal definition of (defect, defect) being a Nash equilibrium. There is no incentive for either individual to cooperate, so how could that strategy have been favored by natural selection?

Current evolutionary thoery relies upon Hamilton's rule and the ideas of inclusive fitness. This rule is $rb - c > 0$, where r is the relation of an organism to another, b is the benefit, and c is the cost. Thus we have that animals will be expected to cooperate when the benefit outweighs the cost, after accounting for the relationship.

We can use evolutionary graph theory to divise another rule where organisms are expected to cooperate. We can generalize the Prisoner's dilemma interaction to analzye the influence of the exact benefit and cost relationship. The following table shows the payoffs for the row player in a general cooperative interaction.

	cooperate	defect
cooperate	b-c	-c
defect	b	0

Where c is the cost of cooperating and b is the benefit. [Ohtsuki et al., 2006] found that the condition $b/c > k$ is necessary for the fixation of cooperation, where k is the average number of neighbors.

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4.4 Numerical Simulations

Numerical simulations were done in a similar manner to [Ohtsuki et al., 2006]. Graph structures were generated using the networkx python package in an Erdős-Rényi manner, but with the added stipulation that the resulting graph be connected [Erdös and Rényi, 1959]. The graphs were populated entirely with defectors, then a node was chosen at random to be a cooperator. A Death-birth with death bias update rule was used at each time step. First, nodes played strategic interactions with all of their neighbors, payoffs were calculated, and fitness values updated accordingly with $w = 0.01$. Then, the node to die was chosen uniformly from the set of all nodes, and it's strategy was replaced by the strategy of one of its neighbors with probability proportional to its fitness. The graph continued to update in this way until cooperators reached fixation or went extinct. Fixation probabilities were calculated by dividing the number of runs that reached fixation by the total number of runs, 10^5. Trials were run with a variety of average connectivity and benefit-to-cost ratios. The average connectivity, k, was tested with values $3, 4, 6, 8, 10$. Nine benefit-to-cost ratios were centered around each value of k. For example, for $k = 3$, b/c was tested at $1, 1.5, 2, 2.5, 3, 3.5, 4, 4.5, 5$.

Results are shown in Figure 6 and support the finding the benefit-to-cost ratio rule is necessary but not sufficient for the evolution of cooperation.

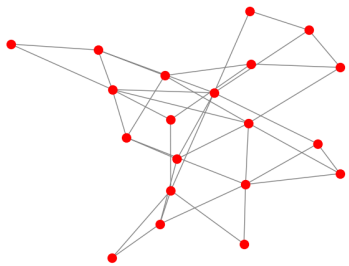

Figure 4: An example of a random connected graph used for simulations.

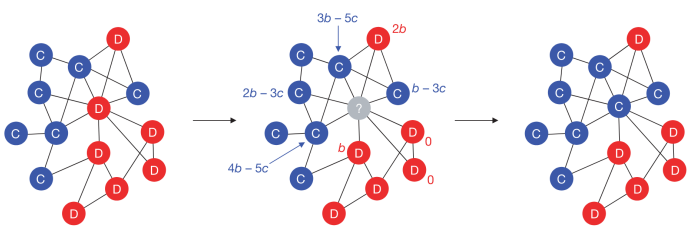

Figure 5: An example of one round of updating. A node is chosen at random to die, payoffs are calculated, and the neighbors are chosen with probability proportional to their fitness to reproduce and replace the node chosed to die. Reproduced from [Ohtsuki et al., 2006].

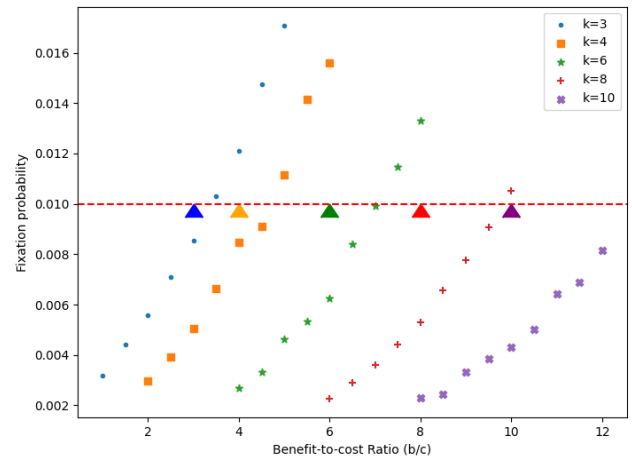

Figure 6: Simulations support b/c $>$ k rule: The graph shows the proportion of attempts out of 10^5 runs that reached fixation, i.e. the number of cooperators reached the total population $N = 100$. The average connectivity is represented using k and the benefit-to-cost ratio is given by the x-axis. The dashed line represents the expected number of runs that would reach fixation under neutral drift, $\frac{1}{N} = 0.01$. The triangles mark $b/c = k$.